Science Olympiad — AddRidge - SSSS

Exam Preparation

You will need:

1. Folders for each of the teams to hold the tests
2. Sufficient copies of the test for all teams. They don't need to be stapled.
3. Multiple timers which have a lap function on them - ideally one per volunteer. The timer app on an iPhone or Android Phone that has a stopwatch function with lap function is sufficient.

Before the event begins:

1. Practice starting the timers and using the lap function to record the times. Make sure volunteers understand how to use the lap function and are not accidentally stopping the timer completely.
2. Memorize the answer to the timed question.
3. Check to make sure that this key matches the test you are proctoring.
4. Place one copy of the test for each team in the provided folders with the first page outside the folder.
5. Adjust desks and chairs – teams may have up to 3 students for this event.

Running the Event

1. When the students enter the room, instruct them to sit down, DO NOT OPEN THE FOLDER, and put their names, school name and school number on the first page.
2. Encourage them to write their team number on all the other pages AFTER they begin the test. This way if their papers gets separated from each other we can make sure to give them credit.
3. **CRITICAL:** Check to see that students have ONLY brought
   i. Something to write with (pencils, pens, erasers)
   ii. Five function calculators (addition, subtraction, multiplication, division, and usually square root). The calculator can have a simple memory store/recall function but must not have a modulus or other scientific and programmable functions. If their calculator doesn't meet these requirements, they may not use it.
   iii. If the student has a smart watch (Apple watch, Samsung Gear, etc.) they will need to put it away.
4. Instruct the students that if they answer the timed question within 10 minutes, they can be awarded a bonus if they solve the timed question with no more than 2 letters incorrect.
   i. When they have a solution for the cryptogram they should raise their hand.
   ii. Let them know that you will announce when the 10-minute time is up. After the first 10 minutes, no additional bonus points will be awarded.
   iii. When you see a team raise their hand, hit the LAP function and head to the team.
   iv. Determine if their answer is correct (see next page for grading), If so, write the time on their score sheet.
   v. If their score is incorrect (more than 2 letters incorrect), tell the team that the answer is wrong, but DO NOT tell them what is wrong. They can continue to work on the question and raise their hand again to be checked. A team has an unlimited number of attempts during the 10-minute bonus.
5. Tell the teams that they do not have to fill in the frequency table. It is simply there as an aid to them solving the cryptogram. It will not be graded.
6. Some students may never have used a non-scientific calculator. You should have them enter a simple formula on their calculator: 1 / 26 = * 26 = .. Most will be surprised to see that the answer is not rounded to 1 as they expected but .9999999999
7. When the timers hit the 10-minute point, announce that no bonus points will be awarded and put away the timers. The students may continue to work on the question, but they may not receive any extra points.

8. A team is not restricted to only the timed question during the 10 minutes. They can move on or split up the work if they would like, but it is in their best interest to try for the bonus.
9. When time is up, have the students put writing instruments down and put their answer pages back into the folder in the correct order.

How to grade

1. Teams can have up to two incorrect letters total on their cryptogram and still be correct. The frequency of the incorrect letter is irrelevant. See the example below.
   If the cryptogram was as shown:
   KZBAOF KFXMFXYF
   SAMPLE SENTENCE

   and the students answered (underlined letters indicate mistakes)
   SAMPL<u>F</u> S<u>F</u>NT<u>F</u>NC<u>F</u>

   then it counts as four mistakes (even though the mistake was only in the letter E) and the answer DOES NOT count. However, if they put
   SAMP<u>UL</u> SENTENCE

   It is considered correct with two letter mistakes.
2. For questions which have a numeric answer (such as determining the a= and b= values), no mistakes are allowed.
3. Teams do NOT have to fill in the frequency table. It is simply there as an aid to them solving the cryptogram. It WILL NOT be graded. It is included in the answer key as an aid to the grader.
4. When scoring the Baconian ciphers (with strange text or symbols), they can write the answer under the Baconian symbols or on the line provided. Note that you will see lots of As and Bs, but they are not graded as the answer, only what they put on the answer line.
5. As you score each question, if correct, put the number of incorrect letters (0, 1, or 2) next to the question number on the scoring page. Also, put the value for the question into the score column. If they get more than 2 letters wrong, subtract 100 points from the score until it would be zero. If a question is worth 240 points and they get 4 letters wrong, you would start with 240 points (for up to 2 letters wrong) and then subtract 100 points for the next two letters wrong ending up with a final score of 40 points for that question. If they had gotten 5 or more letters wrong on a 240 point question, they would receive 0 points for that question. With a 650 point question, they could get 8 letters wrong and receive 50 points (2 free letters then 6×100=600 points off). Just put the incorrect cost deduction on the score sheet and subtract it from the value for the question. Under no circumstance should the score for any question be less than zero. Note that while the timed question must have 2 or fewer letters incorrect in order to get the timing bonus, a team solving the timed question after the 10 minutes passed would be accepted as correct with 3 incorrect letters receiving 100 points for the timed question.
6. If they correctly answered the timed question in 10-minutes or less with 2 or fewer letters incorrect, you need to compute the bonus time. Take the value for the minute from this first table below

0:xx	2,160	1:xx	1,920	2:xx	1,680	3:xx	1,440	4:xx	1,200
5:xx	960	6:xx	720	7:xx	480	8:xx	240	9:xx	0

and then add the seconds value from this table:

X:00	240	X:01	236	X:02	232	X:03	228	X:04	224	X:05	220
X:06	216	X:07	212	X:08	208	X:09	204	X:10	200	X:11	196
X:12	192	X:13	188	X:14	184	X:15	180	X:16	176	X:17	172
X:18	168	X:19	164	X:20	160	X:21	156	X:22	152	X:23	148
X:24	144	X:25	140	X:26	136	X:27	132	X:28	128	X:29	124
X:30	120	X:31	116	X:32	112	X:33	108	X:34	104	X:35	100

X:36	96
X:42	72
X:48	48
X:54	24

X:37	92
X:43	68
X:49	44
X:55	20

X:38	88
X:44	64
X:50	40
X:56	16

X:39	84
X:45	60
X:51	36
X:57	12

X:40	80
X:46	56
X:52	32
X:58	8

X:41	76
X:47	52
X:53	28
X:59	4

For example if they solved the time question at the 6:46 mark, you would add 720 (from the 6:xx entry in the first table) to 56 (from the X:46 entry in the second table) to get a bonus of 776. If they had solved it in exactly 4:00 minutes, you would add 1200 and 240 to get a bonus of 1440.

7. Add up all the scores and put the total on the bottom of score sheet.
8. You must break all ties. Indicate the tie breaker by adding .1 to the score of the team ahead. With multiple teams tied, you will add more. I.e. if five teams all scored 200 points, the final scores that you would enter on the score sheet would be 200.4, 200.3, 200.2, 200.1 and 200.
9. To determine how to break the tie, you need to look at the correctly answered questions in the order from the table below. If both teams answered the same (i.e. they answered the question with zero mistakes) then you go on to the next question. If one team had no mistakes and the other team had one mistake, then the team with no mistakes is ahead. For example, if one team answered question #8 (which is the highest value question) and another team didn't, the first team will be ahead.

Tie Breaker Order	Question #
1	7
2	23
3	9
4	8
5	10
6	6
7	5
8	21
9	4
10	20
11	16
12	15
13	22
14	19
15	18
16	17
17	14
18	3
19	1
20	2
21	Timed
22	13
23	12
24	11

0. If there is still a tie (typically when you have teams which answered either zero, one or two questions) then you will need to look at the tie breaker questions again and count the number of correctly answered letters. The team with the most correctly matched letters is to be ahead.

Timed Question **[200 points]** Solve this timed aristocrat. When you have solved it, raise your hand so that the time can be recorded and the solution checked.

LJ KJZZRHRC KOWR DJWR OIJVY NJU OKKVWOYRZT TJV KOS
DO COLLEGES CARE MORE ABOUT HOW ACCURATELY YOU CAN

OSCURW XVRCYPJSC OIJVY WOSLJD BOCCOHRC, JW UNOY TJV
ANSWER QUESTIONS ABOUT RANDOM PASSAGES, OR WHAT YOU

KOS YWVZT LJ?
CAN TRULY DO?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		1	7	2				2	2	12	6	3		2	12	1		8	5	4	3	7	7	1	6	4
Replacement	Z	P	S	M	K	V	X	G	B	O	C	D	F	H	A	I	J	E	N	Y	W	U	R	Q	T	L

1) **[250 points]** Solve this aristocrat where the ciphertext Z maps to plaintext D.

DXDKOTIZO UEGPDZ PI WGIU UVEP C UINJZ ZI CL C ZCZG'P
EVERYBODY WANTED TO KNOW WHAT I WOULD DO IF I DIDN'T

UCG. C ANDQQ UD'JJ GDXDK WGIU.
WIN. I GUESS WE'LL NEVER KNOW.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1		6	7	2		6		6	3	2	1		2	2	4	2			1	7	1	2	2		6
Replacement	G	Q	I	E	A	X	N	J	O	L	R	F	P	U	Y	T	S	C	Z	B	W	H	K	V	M	D

2) **[200 points]** Decode this quote about a cereal.

YL ADO CWKJQ CIB OPQYPE, Y ADYPT Y CWFJQ EKIX BWNO
IF THE WORLD WAS ENDING, I THINK I WOULD GRAB SOME

RYPPINWP AWIBA RKFPRD, BWNO CIAOK, IPQ Y'Q SKWXIXJG
CINNAMON TOAST CRUNCH, SOME WATER, AND I'D PROBABLY

EW RKIVG.
GO CRAZY.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	5	4	4	3	3	2	2		8	3	6	1		3	5	8	5	4	1	1		1	8	3	7	
Replacement	T	S	W	H	G	U	Y	J	A	L	R	F	Q	M	E	N	D	C	P	K	V	Z	O	B	I	X

3) **[250 points]** Solve this K1 aristocrat which is a quote from Nikola Tesla.

XNL OVLMNTW ZYU XNL CZVHVYDW ZLT VYWTJZLZAHT, HVGT
OUR VIRTUES AND OUR FAILINGS ARE INSEPARABLE, LIKE

CXLBT ZYU IZMMTL. PETY METR WTJZLZMT, IZY VW YX IXLT.
FORCE AND MATTER. WHEN THEY SEPARATE, MAN IS NO MORE.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	1	2	1	2		1	3	3	2		9	5	3	1	1		1		12	2	6	5	5	7	10
Replacement	B	C	F	G	H	J	K	L	M	P	Q	R	T	U	V	W	X	Y	Z	E	D	I	S	O	N	A

4) **[350 points]** Solve this K2 aristocrat about chaos theory.

VFBPTPLBVFSYI, K APFODYDZ NRFLVLBVFS RQ BGR TROL VL
INTERESTINGLY, A PENDULUM CONSISTING OF TWO RODS IS

NRFLVOPTPO K OIFKZVN LILBPZ GVBU NUKRBVN LRYDBVRFL.
CONSIDERED A DYNAMIC SYSTEM WITH CHAOTIC SOLUTIONS.

Replacement	K	M	N	O	P	Q	S	U	V	W	X	Y	Z	F	R	A	C	T	L	B	D	E	G	H	I	J
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	8		3		8	2		3		4	10		5	5	6	1	8	2	3	2	10			3	3

5) **[400 points]** Solve this aristocrat that is a quote from comedian Jack Handey.

GSDRAS ZRW IAQOQIQUS KRYSRLS, ZRW KJRWPB FHPN H YQPS
BEFORE YOU CRITICIZE SOMEONE, YOU SHOULD WALK A MILE

QL OJSQA KJRSK. OJHO FHZ FJSL ZRW IAQOQIQUS OJSY,
IN THEIR SHOES. THAT WAY WHEN YOU CRITICIZE THEM,

ZRW HAS H YQPS HFHZ DARY OJSY HLB ZRW JHXS OJSQA
YOU ARE A MILE AWAY FROM THEM AND YOU HAVE THEIR

KJRSK.
SHOES.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	7	2		2		4	1	10	4	10	6	4		1	8	4	11	12	17		2		6	1	6	7
Replacement	R	D	P	F	X	W	B	A	C	H	S	N	G	K	T	L	I	O	E	Q	Z	J	U	V	M	Y

6) **[450 points]** Solve this aristocrat that has been misspelled quite badly.

OYLOO ET EMKOB FOOFIG ZKENZ KEN MKOR UBO IYKUFFOO.
ENVEE OF OTHER PEEPUL SHOWS HOW THEY ARE UNHAPPEE.

MKOQB XEYMQYIUG UMMOYZKIY ME EMKIBZ DOKULQEB ZKEZ
THEIR CONTINUAL ATTENSHUN TO OTHURS BEHAVIOR SHOS

KEN MKOR UBO DENBQYH.
HOW THEY ARE BOWRING.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency		7		2	11	4	2	1	5		12	2	9	4	15		4	2		1	6			1	7	6
Replacement	K	R	X	B	O	P	L	G	U	Z	H	V	T	W	E	D	I	Y	M	F	A	J	Q	C	N	S

7) **[650 points]** Solve this patristocrat.

FDQFW PDQJF WADFD YDFCW ACNCY HUJOC VHBVZ BHJXC
ITHIN KTHEI NSTIT UTION SOFOU RDEMO CRACY AREWO

WUJHN YRDQF WEADQ BDHFE QDWCX BHJKC KYRBD JUMZA
NDERF ULTHI NGSTH ATRIG HTNOW AREPO PULAT EDBYS

COJDJ HHFMR JKJCK RJ
OMETE RRIBL EPEOP LE

I think the institutions of our democracy are wonderful things that right now are populated by some terrible people.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	5	9	11	2	8		8		11	4		2	2	2	1	5	4			3	2	6	2	4	2
Replacement	S	A	O	T	G	I	Z	R	Q	E	P	J	B	F	M	K	H	L	V	X	D	C	N	W	U	Y

8) **[550 points]** Solve this K1 Patristocrat that is a quote from inventor Dean Kamen.

KGKDO PAROT NJEAP KNPXK EABND EUQUB NEKUZ NSFEF
EVERY BODYH ASTOB EABLE TOPAR TICIP ATEIN AFUTU

DKETN EETKO MNZEE AXUGK SADET NEJMT NEEKQ TZAXA
RETHA TTHEY WANTT OLIVE FORTH ATSWH ATTEC HNOLO

LOQNZ RA
GYCAN DO

Everybody has to be able to participate in a future that they want to live for. That's what technology can do.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	8	2		4	14	2	2			2	9	1	2	10	4	3	3	2	2	6	4			3		4
Replacement	O	P	Q	R	T	U	V	X	Z	S	E	G	W	A	Y	B	C	D	F	H	I	J	K	L	M	N

9) **[550 points]** Solve this patristocrat. You are told it starts with "Some".

NZLTZ JLKLZ NHSLW ZUHFR HLENS IFBTO WTUST RITNY
SOMEO FMYMO STIMP ORTAN TMUSI CALEX PERIE NCESW

TUTJU ZLFME URHIG YSHDN ZRCNL KJUST RGGZY RBZFG
EREFR OMABU RNTCD WITHS ONGSM YFRIE NDDOW NLOAD

TGJZU LTFHF HTUUS MBTGS CSHFB AEFBS HK
EDFOR MEATA TERRI BLEDI GITAL QUALI TY

Some of my most important musical experiences were from a burnt CD with songs my friend downloaded for me at a terrible digital quality.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	5	2	1	3	8	6	9	3	4	3	8	2	6	1			6	9	12	9		2		3	9
Replacement	Q	L	G	H	U	A	D	T	C	F	Y	M	B	S	X	Z	J	N	I	E	R	K	P	V	W	O

10) **[500 points]** Solve this K1 xenocrypt that has been encoded with an English keyword.

VWUXJBEILY DVAX IÑADLY VKLY MÑX HLY UEALY WÑXBXJ
APRENDIMOS HACE MUCHOS AÑOS QUE LOS RICOS PUEDEN

NXJXU BEJXUL, WXUL HLY WLZUXY NEXJXJ NEXIWL.
TENER DINERO, PERO LOS POBRES TIENEN TIEMPO.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	3		2	5			2	3	6	1	10	1	3	3							6	3	5	13	7	1
Replacement	C	D	F	H	I	J	K	L	M	N	Ñ	O	Q	T	U	V	W	X	Y	Z	G	R	A	P	E	S	B

11) **[100 points]** Solve this caesar cipher.

Q	N		Q		K	W	C	T	L		Z	M	U	M	U	J	M	Z		B	P	M		V	I	U	M	A		W	N		I	T	T
I	**F**		**I**		**C**	**O**	**U**	**L**	**D**		**R**	**E**	**M**	**E**	**M**	**B**	**E**	**R**		**T**	**H**	**E**		**N**	**A**	**M**	**E**	**S**		**O**	**F**		**A**	**L**	**L**

B	P	M	A	M		X	I	Z	B	Q	K	T	M	A	,		Q	'	L		J	M		I		J	W	B	I	V	Q	A	B	.
T	**H**	**E**	**S**	**E**		**P**	**A**	**R**	**T**	**I**	**C**	**L**	**E**	**S**	,		**I**	'	**D**		**B**	**E**		**A**		**B**	**O**	**T**	**A**	**N**	**I**	**S**	**T**	.

12) **[100 points]** Solve this Caesar cipher that is a quote about salad.

S	L	G	T	Y	R		N	L	P	D	L	C		D	L	W	L	O		H	T	E	S	Z	F	E		N	C	Z	F	E	Z	Y	D		T	D
H	**A**	**V**	**I**	**N**	**G**		**C**	**A**	**E**	**S**	**A**	**R**		**S**	**A**	**L**	**A**	**D**		**W**	**I**	**T**	**H**	**O**	**U**	**T**		**C**	**R**	**O**	**U**	**T**	**O**	**N**	**S**		**I**	**S**

W	T	V	P		S	L	G	T	Y	R		X	L	N		L	Y	O		N	S	P	P	D	P		H	T	E	S	Z	F	E		E	S	P
L	**I**	**K**	**E**		**H**	**A**	**V**	**I**	**N**	**G**		**M**	**A**	**C**		**A**	**N**	**D**		**C**	**H**	**E**	**E**	**S**	**E**		**W**	**I**	**T**	**H**	**O**	**U**	**T**		**T**	**H**	**E**

N	S	P	P	D	P	.
C	**H**	**E**	**E**	**S**	**E**	.

13) **[150 points]** Encode this phrase given that a = 15 and b = 7.

P	R	O	G	R	A	M	M	I	N	G		I	S	N	'	T		A	B	O	U	T		W	H	A	T		Y	O	U		K	N	O	W	;
Y	**C**	**J**	**T**	**C**	**H**	**F**	**F**	**X**	**U**	**T**		**X**	**R**	**U**	**'**	**G**		**H**	**W**	**J**	**V**	**G**		**Z**	**I**	**H**	**G**		**D**	**J**	**V**		**B**	**U**	**J**	**Z**	;

I	T	'	S		A	B	O	U	T		W	H	A	T		Y	O	U		C	A	N		F	I	G	U	R	E		O	U	T
X	**G**	**'**	**R**		**H**	**W**	**J**	**V**	**G**		**Z**	**I**	**H**	**G**		**D**	**J**	**V**		**L**	**H**	**U**		**E**	**X**	**T**	**V**	**C**	**P**		**J**	**V**	**G**

14) **[250 points]** Solve this affine cipher that ends with "LS", and is a quote from Kawhi Leonard.

V	'	I	J		X	E	L	X	R	Z		E	V	B	J	G		S	N	H	J	-	D	N	N	B	J	G		H	J	X	E	Z	.
I	**'**	**V**	**E**		**A**	**L**	**W**	**A**	**Y**	**S**		**L**	**I**	**K**	**E**	**D**		**H**	**O**	**M**	**E**	-	**C**	**O**	**O**	**K**	**E**	**D**		**M**	**E**	**A**	**L**	**S**	.

15) **[300 points]** Encode this 3 by 3 Hill with keyword 'Aardvarks'.

$$\begin{pmatrix} A & A & R \\ D & V & A \\ R & K & S \end{pmatrix} \equiv \begin{pmatrix} 0 & 0 & 17 \\ 3 & 21 & 0 \\ 17 & 10 & 18 \end{pmatrix}$$

A	R	T	H	U	R	T	I	M	O	T	H	Y	R	E	A	D	
L	**T**	**S**	**D**	**Z**	**B**	**W**	**R**	**V**	**P**	**Z**	**I**	**Q**	**N**	**A**	**J**	**L**	**M**

16) **[300 points]** Decode this hill cipher.

$$\begin{pmatrix} H & O \\ W & D \end{pmatrix} \equiv \begin{pmatrix} 7 & 14 \\ 22 & 3 \end{pmatrix}$$

I	L	Y	H	A	N	I	N	U	M
A	**V**	**A**	**L**	**A**	**N**	**C**	**H**	**E**	**S**

17) **[250 points]** Decode this Vigenere cipher, whose keyword is 'Creed'.

C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R	E	E	D
K	M	I	F	H	G	E	M	R	Y	Q	C	Z	I	G	K	E	E	R	X	O	S	I	V	R	H	T	Y	P	W
I	**V**	**E**	**B**	**E**	**E**	**N**	**I**	**N**	**V**	**O**	**L**	**V**	**E**	**D**	**I**	**N**	**A**	**N**	**U**	**M**	**B**	**E**	**R**	**O**	**F**	**C**	**U**	**L**	**T**

C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R	E	E	D
U	S	S	X	K	C	C	I	E	G	G	I	E	R	G	C	W	S	P	O	Q	N	I	V	B	Q	L	L	E	Y
S	**B**	**O**	**T**	**H**	**A**	**L**	**E**	**A**	**D**	**E**	**R**	**A**	**N**	**D**	**A**	**F**	**O**	**L**	**L**	**O**	**W**	**E**	**R**	**Y**	**O**	**U**	**H**	**A**	**V**

C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R	E	E	D
G	D	S	V	H	H	L	R	E	V	C	W	S	P	O	Q	N	I	V	E	W	K	C	S	X	O	R	O	I	P
E	**M**	**O**	**R**	**E**	**F**	**U**	**N**	**A**	**S**	**A**	**F**	**O**	**L**	**L**	**O**	**W**	**E**	**R**	**B**	**U**	**T**	**Y**	**O**	**U**	**M**	**A**	**K**	**E**	**M**

C	R	E	E	D	C	R	E	E	D	C	R	E	E	D	C	R
Q	I	I	Q	R	P	V	C	E	V	C	C	I	E	G	G	I
O	**R**	**E**	**M**	**O**	**N**	**E**	**Y**	**A**	**S**	**A**	**L**	**E**	**A**	**D**	**E**	**R**

18) **[250 points]** Decode this Vigenere that starts with "Learn From"

E	I	N	S	T	E	I	N	E	I	N	S	T	E	I	N	E	I	N	S	T	E	I	N	E	I	N	S	T	E
P	M	N	J	G	J	Z	B	Q	G	R	K	M	I	Z	Q	E	G	Y	A	O	I	N	B	V	B	B	V	T	C
L	**E**	**A**	**R**	**N**	**F**	**R**	**O**	**M**	**Y**	**E**	**S**	**T**	**E**	**R**	**D**	**A**	**Y**	**L**	**I**	**V**	**E**	**F**	**O**	**R**	**T**	**O**	**D**	**A**	**Y**

I	N	E	I	N	S	T	E	I	N	E	I	N	S	T	E	I	N	E	I	N	S	T	E	I	N	E	I	N	S
P	B	T	M	S	G	K	X	W	Z	S	Z	E	G	P	X	P	R	M	U	C	G	K	X	I	A	X	B	U	A
H	**O**	**P**	**E**	**F**	**O**	**R**	**T**	**O**	**M**	**O**	**R**	**R**	**O**	**W**	**T**	**H**	**E**	**I**	**M**	**P**	**O**	**R**	**T**	**A**	**N**	**T**	**T**	**H**	**I**

T	E	I	N	E	I	N	S	T	E	I	N	E	I	N	S	T	E	I	N	E	I	N	S
G	K	Q	F	R	W	G	L	H	W	B	B	T	Y	H	W	L	X	Q	B	R	Q	A	Y
N	**G**	**I**	**S**	**N**	**O**	**T**	**T**	**O**	**S**	**T**	**O**	**P**	**Q**	**U**	**E**	**S**	**T**	**I**	**O**	**N**	**I**	**N**	**G**

19) **[250 points]** Solve this Morbit cipher where 1 = '••', 2 = --, 9 = 'x•', 8 = 'x-', 4 = '-•', and 5 = '•-'.

```
4 9 5 8 6 3 5 3 2 6 1 6 1 3 9 6 5 1 8 4 8 2 9 4 9 3
-●x●●-x--x●x●-●x---x●●-x●●●xx●-x●-●●x--●x---x●-●x●●x
N  U   M  E R   O   U   S   /A  L    G   O   R   I

6 1 1 8 6 1 3 8 5 3 5 8 3 8 1 3 3 9 5 9 1 9 8 1 7 6
-x●●●●x--x●●●xx-●-●x●-x-●xx-●●●x●xx●●-x●●●x●x-●●xx-x
T H    M  S   /C    A  N  /B    E /U   S   E D  / T

2 6 9 1 8 2 9 4 3 1 5 9 7 6 1 1 9 7 6 5 3 5 9 1 6
---xx●●●x---x●-●●x●●●-x●xx-x●●●●x●xx-x●-●x●-x●●●-x
O   /S   O   L    V    E/ T H    E/ T R   A  V

3 5 1 9 3 4 8 4 7 1 3 5 9 4 3 3 1 3 2 9 6 4 7 5 4
●x●-●●x●●x-●x--●xx●●●x●-x●-●●x●x●●●x--x●-x-●xx●--●
E L    I  N  G  / S   A  L    E S   M  A  N / P

9 4 8 2 8 1 3 5 1 9 8 6
x●-●x---x-●●●x●-●●x●x--x
 R   O   B    L    E M
```

20) **[300 points]** Decipher this quote encoded using the Morbit cipher. You are told it ends with "CRAZY" and that 6 maps to '-x'.

```
7 9 5 6 2 6 7 6 5 7 8 7 9 7 8 6 5 8 4 1 3 4 1 9 6
●●xx●--x---x●●-x●-●●x-●●xx●●x--x●-x--●x●●x-●x●xx-x
I / W   O   U   L    D  / I  M  A  G   I  N  E/ T

7 7 1 6 6 1 3 7 4 9 4 2 8 2 1 5 9 4 4 8 2 1 5 1 4 3
●●●●x●-x-xx●●x●●-●xx-●--x---x●●-xx-●-●x---x●●-x●-●●x
H    A  T /I  F   / Y    O   U  / C    O   U   L

4 3 1 5 8 3 4 3 3 5 3 7 3 6 5 8 3 4 3 8 6 2 6 5 3
-●●xx●●-x-●x-●●x●x●-●x●●●x-x●-x-●x-●●xx--x---x●-●x
D   /U   N  D   E R   S   T A  N  D   /M  O   R

7 3 3 8 5 3 2 6 4 3 3 1 6 8 1 6 5 4 9 4 3 5 8 3 4 4
●●●x●xx-●-●x---x-●●x●xx●-xx-x●-x●--●xx-●●x●-x-●x-●-●
S   E /C    O   D   E /A  /T A  P   / D   A  N  C

1 1 4 9 5 6 2 6 7 6 5 7 8 7 9 4 3 5 3 7 1 7 6 3 8 5 6
x●x●-●xx●--x---x●●-x●-●●x-●●xx-●●x●-●x●●x●●●-x●xx-●--x
 E R  / W   O   U   L    D  / D   R   I  V    E /Y

2 6 7 6 8 5 3 5 3 5 8 4 3 4 2
---x●●-xx-●-●x●-●x●-x--●●x-●--
O   U   /C    R   A  Z    Y
```

21) **[350 points]** Decode this Pollux cipher that is a quote from Tyler, The Creator given that 1 = x, 2 = •, 5 = -, 7 = -, 9 = x, 0 = x

```
22693280555932903668824685122251890870051428494947 4185
●●xx-●●x---x-●xx-xx●●●●x●-x●●●-x●xx●-xx-x●●●●x●x●-●x●-
I   D   O   N   T  H    A  V    E  A   T H    E R   A

68334988024813092429733102899883048492607504612304226053
x●--●x●●x●●●x-xx●●●x---xx●●xx●●-x●●●x●xx--x●xx●-x●●●xx--
 P    I  S   T  S   O    I   U   S   E  M  E  A  S    M

63473113339833032115982841802720859473418808440300873
x-●--xx---x●--x-●xx-x●●●●x●x●-●x●-x●--●x●●x●●●x-xx●--
 Y     O   W   N   T H    E R   A  P    I  S   T  W

94448981526142913799536256725988054177216514284121153
x●●●●x●x-●xx●●xx--xx--x●-x-●-x●●x-●x--●xx-x●●●●x●xx--
 H    E N   I   M   M  A  K   I  N  G    T H    E  M

9487022418867838
x●●-x●●●x●●x-●-●
 U   S   I  C
```

22) **[250 points]** Solve this tongue twister encoded as a Baconian cipher.

```
^!!!^!^!!!^!^!^^!!!^!^!^!!!^!!!!^!!!^!!^^!!!^^!^!!!!!
BAAABABAAABABABBAAABABABAAABAAAABAAABAABBAAABBABAAAAA
  S    I    X    S    L    E    E    K    S    W    A

!!!^^!!^!!!^^!!!^^!^!!!!!!!!!^!^^^!!!^^!^!!!^!!!!!^!^^
AAABBAABAAABBAAABBABAAAAAAAAABABBBAAABBABAAABAAAAABABB
    N    S    S    W    A    M    S    W    I    F

!!^!!^!^!^!^^!^!!!^!^^!^^!!^^^!!^!!!^^^^!^!!!!!!!^!!!
AABAABABABABBABAAABABBABBAABBBAABAAABBBBABAAAAAAABAAA
 T    L    Y    S    O    U    T    H    W    A    R

!!!!^^^!!!^
AAAABBBAAAB
   D    S
```

Six sleek swans swam swiftly southwards.

23) **[600 points]** Megan, has faithfully followed the steps of the RSA key-generation algorithm. Here are the results:

$p = 341$

$q = 623$

$n = 232481$

$\Phi = 341271$

$e = 235663$

Unfortunately, Megan doesn't know how to compute the value of *d* and needs you to do that final step for them.

Enter the computed value of *d*, NOT the formula.

263119